MKOGP3
Cognitive Modeling

Annotated Reading List VL's 2–5

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There is substantial heterogeneity in most cohorts of the Master in Cognitive Science at the University of Tübingen: Some come with substantial formal training, e.g. with a Bachelor in mathematics, computer science or physics, some with substantial knowledge about cognition, e.g. with a Bachelor in psychology or cognitive science, some, finally, with substantial understanding of neuroscience, e.g. with a Bachelor in biology. The core courses—in German *Grundlagen (Pflicht)* in the *Modulhandbuch*—are designed to ensure that everyone, regardless of their Bachelor, acquires a minimal set of common basic knowledge, skills and formal training. This lecture course *Cognitive Modeling* is one of the core courses. Thus, for some of you with a strong formal background much of what is covered in this course may be easy, perhaps even trivial. Others, however, may find the course rather taxing, requiring you to read book chapters and original research papers in a style unusual to you, and, on top, to learn to programme in R.[1] This it is particularly true for the first four content lectures, VL2 to VL5, as they cover *modeling fundamentals*—in my experience there is an approximately uniform distribution of students with respect to how difficult they find the four modeling fundamental lectures: From trivial to nearly impossible. There is only one solution I see: Help each other! If you find the course easy, please help your colleagues who struggle. If you are in trouble, go and seek out one of the formal wizards in your cohort to explain things to you.

[1] Sometimes this may lead to frustration for some of you. During those times it may help to remember, however, that this is inevitable in a Master programme that allows students with diverse backgrounds to enrol in it.

As announced in the first lecture (VL1) at least one of your lecturers—that is, me, Felix Wichmann—will run his lectures *partially* in an *inverted classroom* style.[2] For the inverted classroom to be useful and enjoyable it is essential that you prepare yourselves *before* our meetings: You will have to read some book chapters and/or original research papers prior to class.

[2] For details please consult the relevant slides in the introductory material uploaded on ILIAS and, if interested, watch the videos by Eric Mazur mentioned in the slides.

For the first three of my lectures on the basics of modeling—estimation (VL2), goodness-of-fit (VL3) and assessing variability (VL4)—only the material covered during the lectures is relevant for the exam: We focus on the statistical and conceptual fundamentals of modeling, not on any details of the specific psychological models—or the R code—used to introduce the statistics and concepts in the textbook we are using, e.g. the "Wald model".

THE FIFTH LECTURE INTRODUCES BAYESIAN STATISTICS to you. The application of Bayesian statistics to psychology, cognitive science and the neurosciences has become rather frequent for at least a decade or two. The same is true for Bayesian models of cognition (or neuroscience), i.e. the idea that the mind and brain may analyse and respond to the world like a Bayesian statistician would do. In the fifth and final lecture on modeling fundamentals I will introduce Bayesian statistics, highlighting which aspects are the same, and which are different from "traditional" or "frequentist" statistics as we talked about in lectures VL2 to VL4.

VL2: RMSD and ML

The second lecture (VL2) is about a fundamentally important topic in modeling data in general, whether cognitive, physical or whatever: about *root mean squared deviation* (RMSD) and *maximum likelihood* (ML).[3]

[3] RMSD is frequently also referred to as *root mean squared error* (RMSE) and a variant without the root is known as *least mean squared error* (LMSE).

Please read chapters 3 and 4 in Farrell and Lewandowsky (2018) *before* the lecture. In chapter 3 only read sections 3.1–3.4 as well as 3.6; section 3.5 discusses the bootstrap to obtain variability estimates and will be part of your reading assignment for the fourth lecture (VL4). Chapter 4 on maximum likelihood is relevant in its entirety (but there is no need to refer back to the details from previously introduced models like the GCM). Furthermore, if you are not familiar with *R* you can safely skip the implementation sections. You should have online access to the entire book on `https://doi.org/10.1017/CBO9781316272503`—you need to log-in via *institutional log-in* and select *University of Tuebingen* and afterwards enter your ZDV/Login-ID credentials (Shibboleth) to gain acess.[4]

[4] You should thank MARTIN BUTZ for his efforts to make the online access possible—he convinced the University Library to pay for this service for our course!

QUESTIONS TO THINK ABOUT while reading:

1. Why minimise the mean squared deviation and not simply the mean deviation or the mean absolute deviation?
2. Why are local minima of the error function so problematic?
3. Why is `Simulated Annealing` less prone to get stuck in local minima than many other optimisation techniques?
4. Make sure you understand and can explain Figure 4.5 on p. 83.
5. Why is it useful to know that the ML estimator is *minimally sufficient*?

6. What does it mean and why is it desirable that the ML estimator is *consistent*?

VL3: Goodness-of-fit

The third lecture (VL3) is about the second of three fundamentally important topics in modeling data: about *goodness-of-fit* or, more generally, *model assessment and selection*. You should read chapter 10 in Farrell and Lewandowsky (2018) before the class. In chapter 10 please do not spend too much time and effort on the mathematical details and Kullback-Leibler (KL) distance in section 10.4, pp. 256–258. Again, if you are not familiar with R you can safely skip over the implementation sections.

One aspect in chapter 10 that does not get enough attention is *cross-validation*. I suggest you watch an episode from Josh Starmer's *StatQuest* videos called *Machine Learning Fundamentals: Cross Validation*.[5] The video is only around 6 minutes long but really very clear on how cross-validation works and how it is used to compare different models.[6] Alternatively, if you are very keen on delving deeper into mathematical details you may want to consult section 7.10 in chapter 7 in Hastie et al. (2009)—but much of the details (and terminology) in this book are well beyond this course.[7]

At the end of the lecture I will provide an outlook on *regularisation*—regularisation is a modern computational technique attempting to trade-off model complexity and goodness-of-fit.[8]

QUESTIONS TO THINK ABOUT while reading:

1. Can you see a connection between over-fitting and *epicycles*?[9]
2. What exactly is meant by the term *bias* on p. 245, referring to the top left panel in Figure 10.3, p. 246? And, in the same figure, why does the variance *in the fits* increase with more complex models?
3. Figure 10.4 shows the bias-variance trade-off and how it shows the model with the right model complexity—what do you need to know to produce a plot like Figure 10.4? Why is this typically impossible in practice, i.e. when you want to model a cognitive process?
4. Why is the *bias* zero for models of order 3 and above in Figure 10.4 but the *training error* continuously decreases for orders up to 6 in Figure 10.5?
5. Cross-validation (CV) is a numerical, simulation based technique for model assessment: How does it trade-off between goodness-of-fit and model complexity?

[5] `https://www.youtube.com/watch?v=fSytzGwwBVw`

[6] But you have to put up with some 15 seconds of bizarre and cheesy singing at the beginning of the video.

[7] In parts the section is somewhat arcane and only applicable in very restricted settings. A PDF of the book is freely available from the first author's webpage.

[8] The example presented during class is from the excellent book by Bishop (2006), section *1.1. Example: Polynomial Curve Fitting*. It is not compulsory to read this section but I've uploaded the book to ILIAS for those who would like to read a bit more about the topic.

[9] If you unsure about what epicylces are please have a quick look at `https://en.wikipedia.org/wiki/Deferent_and_epicycle`. Alternatively, read section 1.1, pp. 3–6 in Farrell and Lewandowsky (2018).

6. What advantages do you see of CV over, e.g. AIC, MDL or NML? What problems do you see?

VL4: Assessing variability

The fourth lecture (VL4) is about the third and final of the three fundamentally important topics in modeling data: about *assessing the variability of the fitted parameters* or, more colloquially, about *error bars*. Please read section 3.5 in chapter 3 of Farrell and Lewandowsky (2018) as well as one of my own papers: Wichmann and Hill (2001b). The paper is concerned with the psychometric function—do not worry about the details of the psychometric function and instead only concentrate on the issue of the bootstrap to obtain error bars.[10]

[10] Only skim-read all the discussion and details about the different *sampling schemes* which are only important for psychometric function connoisseurs.

QUESTIONS TO THINK ABOUT while reading:

1. Why is it essential to obtain estimates of the variability of the fitted parameters when modeling data?
2. What is the difference between the *non-parametric* and the *parametric* bootstrap?
3. When is it appropriate to use the *non-parametric* bootstrap? When would you use the *parametric* bootstrap?
4. Wichmann and Hill (2001b) write about the *bootstrap bridging assumption*—what is this and why and when do they argue it is required to be tested?
5. What (potential) (parametric) bootstrap problem is illustrated in Figure 6 on p. 1324 in the paper by Wichmann and Hill (2001b)?
6. In their conclusions Wichmann and Hill (2001b) list a number variables influencing the size of the confidence intervals calculated from the parametric bootstrap—which of the crucial variables are *features* and which are *bugs*?

An example of the three steps of modeling data in one single, coherent application is provided by the two companion papers by myself, Wichmann and Hill (2001a,b): Estimating the parameters (VL2), assessing goodness-of-fit and model comparison (VL3), and estimating the variability of the fitted parameters (VL4). As model it uses a somewhat pedestrian example, namely the *psychometric function*. The psychometric function is only a descriptive model—rather than a true cognitive model where we attach meaning to the functional form of the model and individual parameters. The psychometric function is very frequently used to analyse experimental data but it is mainly intended to summarise data in a compact form. Please note that it is not compulsory to read Wichmann and Hill (2001a)—this is only a suggestion for those of you who would like to see a complete modeling "pipeline" as described in the three lectures VL2–VL4.

VL5: Bayesian Statistics

For the fifth lecture (VL5) there is *no* compulsory reading to be done. What is and is not meant or implied by "Bayes" in cognitive science and psychology is sometimes unclear or even confusing. The goal of the fifth lecture is to provide you with an overview of the many issues that are often subsumed under "Bayes", "Bayesian statistics" or "Bayesian modelling", and to clarify which of the issues are potentially contentious.

We will cover, first, *Bayes theorem* which is entirely uncontroversial and accepted by any statistician. A highly contentious issue, however, is the question of what probabilities are; here standard—aka frequentist aka orthodox—probabilities and *Bayesian probabilities* differ substantially (see e.g. the relevant chapter in Dienes, 2008). A third issue is to do *Bayesian statistics* for inference, an alternative way to analyse data and avoid having to calculate p-values (see e.g. Etz et al., 2018; Etz and Vandekerckhove, 2018; Wagenmakers et al., 2018)—we will cover this aspect only in passing. Fourth and importantly for our course on Cognitive Modeling is the issue of *Bayesian modelling*: How does Bayesian modelling differ from the standard (frequentist, orthodox) computational modelling we explored in lectures two, three and four (e.g. Lee and Wagenmakers, 2013; Schütt et al., 2016)? Fifth and finally we look at a controversial issue again, the issue of the *Bayesian mind* or *Bayesian brain*: Does the mind (brain) use Bayesian methods to perceive, memorise or act in the world? Are Bayesian methods thus not only a way to analyse or model data in science, but a theory of how the mind (brain) works? Some cognitive scientists are convinced that the mind (brain) is Bayesian (e.g. Mamassian et al., 2002; Yuille and Kersten, 2006; Ma et al., 2006; Körding and Wolpert, 2006; Zednik and Jäkel, 2016) others argue that this is not true and even a distraction for the field (Bowers and Davis, 2012).

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